

Model Question Paper-II with effect from 2025-26

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1BMATS201

Second Semester B.E./B.Tech. Degree Examination NUMERICAL METHODS: FOR CSE STREAM

TIME: 03Hours

Max.Marks:100

Note: 1. Answer any FIVE full questions, choosing at least ONE question from each MODULE.

2. VTU Formula Hand Book is Permitted

3. M: Marks, L: Bloom's level, C: Course outcomes.

Module-1			M	L	C
Q 1.	a	Define the following i) Round off Error, ii) Inherent error, iii) Truncation error, iv) Absolute Error, v) Relative error, vi) Percentage error	6	L2	CO-1
	b	Obtain a second degree polynomial approximate to $f(x) = (1+x)^{\frac{1}{2}}$, $x \in [0, 0.1]$ using the Taylor series expansion about $x = 0$. Use the expansion to approximate $f(0.05)$ and find a bound of the truncation error.	7	L3	CO-1
	c	A real root of the equation $f(x) = x^3 - 5x + 1 = 0$ lies in the interval (0,1). Perform four iterations of the Regular-Falsi method to obtain this root.	7	L3	CO-1
OR					
Q 2.	a	Find the absolute error if the number 0.00545828 is i) truncated to 3 decimal digits ii) round off to 3 decimal digits	6	L2	CO-1
	b	Find the root of the equation $x^3 - 4x - 9 = 0$, using the bisection method correct to three decimal places.	7	L3	CO-1
	c	Find the real root of the equation $3x = \cos x + 1$ by Newton Raphson method.	7	L3	CO-1
Module-2					
Q 3.	a	Find the condition number of an invertible matrix $A = \begin{bmatrix} 1 & 3 \\ 3 & 5 \end{bmatrix}$ with respect to L_1 norm.	6	L2	CO-1
	b	Apply LU decomposition to find the solution of the system of linear equations $3x + 2y + 7z = 4$; $2x + 3y + z = 5$; $3x + 4y + z = 7$.	7	L3	CO-1
	c	Using Rayleigh's power method, find the dominant eigen value and corresponding eigen vector for the matrix $\begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix}$ by choosing the initial vector $[1 \ 0 \ 0]^T$. Perform four iterations.	7	L3	CO-1
OR					
Q 4.	a	Show that following system of linear equation is ill conditioned. $2x + y = 4$; $2x + 1.01y = 4$	6	L2	CO-1
	b	Use Gauss Seidel method to approximate the solution of system of linear equations $10x + 2y + z = 9$; $2x + 20y - 2z = -44$; $-2x + 3y + 10z = 22$ near the point (0,0,0). Perform three iterations.	7	L3	CO-1
	c	Using Jacobi's method find an approximate eigen values and eigen vectors of the matrix $\begin{bmatrix} 7 & 1 \\ 1 & 7 \end{bmatrix}$.	7	L3	CO-1

Module-3										
Q 5.	a	From the following table, estimate the number of students who obtained marks between 40 and 45.					6	L3	CO-1	
	Marks :	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80				
	No. of Students:	31	42	51	35	31				
	b	Find the piecewise quadratic interpolation polynomial for the function $f(x)$ defined for the data					7	L3	CO-1	
	x	-2	0	1	3	4				
$f(x)$	-23	1	4	80	193					
Hence interpolate at $x = -0.5$ and $x = 2$.										
c	Find the polynomial $f(x)$ by using Lagrange's formula and hence find $f(3)$ for					7	L3	CO-1		
x :	0	1	2	5						
$f(x)$:	2	3	12	147						
OR										
Q 6.	a	Estimate the value of $f(42)$ using appropriate interpolation formula from the following data.					6	L3	CO-1	
	x :	20	25	30	35	40				45
	$f(x)$:	354	332	291	260	231				204
	b	Evaluate $f(9)$, using Newton's divide difference formula, from the data					7	L3	CO-1	
	x :	5	7	11	13	17				
	$f(x)$:	150	392	1452	2366	5202				
	c	Using piecewise linear interpolation formula to find $f(3)$ and $f(1.5)$ from the data.					7	L3	CO-1	
x :	1	2	4	8						
$f(x)$:	3	7	21	73						
Module-4										
Q 7.	a	Solve $(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0$					6	L3	CO-2	
	b	Solve $(D^4 + 4D^3 - 5D^2 - 36D - 36)y = 0$					7	L3	CO-2	
	c	Solve $(D^3 + 2D^2 + D)y = e^{-x} + \sin 2x$					7	L3	CO-2	
OR										
Q 8.	a	Solve $(x^3y^2 + xy)dx = dy$					6	L3	CO-2	
	b	Solve $\frac{d^4y}{dx^4} + 8\frac{d^2y}{dx^2} + 16y = 0$					7	L3	CO-2	
	c	Solve $(D^3 + 1)y = \cos(2x - 1) + 3^x$.					7	L3	CO-2	
Module-5										
Q 9.	a	Evaluate $\int_4^{5.2} \log x \, dx$ by 6 equal parts by Simpson's $1/3^{\text{rd}}$ rule					6	L3	CO-2	
	b	Use Modified Euler's method to compute $y(1.2)$ correct to five decimal places given that $\frac{dy}{dx} + \frac{y}{x} = \frac{1}{x^2}$ and $y = 1$ at $x = 1$, taking $h = 0.1$.					7	L3	CO-2	
	c	Using Milne's method, compute y at $x=0.8$ from $\frac{dy}{dx} = x - y^2$ with the data $y(0) = 0, y(0.2) = 0.02, y(0.4) = 0.0795, y(0.6) = 0.1762$.					7	L3	CO-2	
OR										
Q 10.	a	Use Simpson's $3/8$ th rule to find $\int_0^{\pi/2} \sqrt{\cos x} \, dx$ by dividing $[0, \pi/2]$ into six equal parts.					6	L3	CO-2	
	b	Solve by Taylor's series method the equation $\frac{dy}{dx} = \log(xy)$ for $y(1.1)$ and $y(1.2)$ given $y(1)=2$.					7	L3	CO-2	
	c	Use forth order Runge-kutta method to solve $\frac{dy}{dx} = \frac{y-x}{(x+y)}, y(0) = 1$, at $x=0.1$ correct to four decimal places.					7	L3	CO-2	

CBCS SCHEME

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Second Semester B.E./B.Tech. Degree Examination, June/July 2026 Numerical Methods : CSE Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.
3. VTU Formula Hand Book is permitted.

Module - 1				M	L	C
Q.1	a.	Define Error, Absolute error, Relative error and Percentage errors.		6	L2	CO1
	b.	Using Newton Raphson method find the real root of $xe^x - 2 = 0$ correct to four decimal places.		7	L3	CO1
	c.	Using Bisection method find the real root of the equation $x^3 - 2x - 5 = 0$. Carry out 4 iterations.		7	L3	CO1
OR						
Q.2	a.	Round off the number 37.46235 to 4 significant figures and compute (i) Absolute error, (ii) Relative error (iii) percentage error		6	L2	CO1
	b.	Find a real root of the equation $x \log_{10} x - 1.2 = 0$ by regula falsi method correct to four decimal places.		7	L3	CO1
	c.	Using Secant method find the real root of the equation $x^3 + x^2 - 1 = 0$. Carry out 4 iterations.		7	L3	CO1
Module - 2						
Q.3	a.	Find the condition number of an invertible matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ with respect to L_1 norm.		5	L3	CO1
	b.	Apply LU Decomposition method to solve $x + y + z = 1$; $3x + y - 3z = 5$; $x - 2y - 5z = 10$		8	L3	CO1
	c.	Find the largest eigen value and eigen vector of the matrix A using Rayleigh's power method by taking the initial vector as $[1,1,1]'$ $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$		7	L3	CO1
OR						
Q.4	a.	Verify whether the following system of linear equations is well poised or not using spectral norm $\begin{bmatrix} 2.1 & 1.8 \\ 6.2 & 5.3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2.1 \\ 6.2 \end{bmatrix}$		6	L2	CO1
	b.	Solve by Gauss Seidel method $20x + y - 2z = 17$, $3x + 20y - z = -18$, $2x - 3y + 20z = 25$. Carry out 5 iterations taking $X' = (1,1,1)$		7	L3	CO1

Q.4	c.	Using Jacobi's method find the approximate eigen values and eigen vectors of $A = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$	7	L3	CO1
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Module - 3

Q.5	a.	From the following table find the number of students who obtained marks between 40 and 45	6	L3	CO1
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Marks	30-40	40-50	50-60	60-70	70-80
No of students	31	42	51	35	31

	b.	Obtain the piecewise linear interpolating polynomial for the data and hence interpolate at $x = 0.5$ and 2.5	7	L3	CO1
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x	0	1	2	3
f(x)	1	2	5	10

	c.	Evaluate $f(3)$ from the following table using Newton's divided interpolation formulae.	7	L3	CO1
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x	0	1	4	8	10
f(x)	-5	-14	-125	-21	355

OR

Q.6	a.	Apply appropriate interpolation formula to find $y(3)$.	6	L3	CO1
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x	0	2	4	6	8	10
y	0	4	56	204	496	980

	b.	Obtain the piecewise quadratic interpolating polynomial for the data and hence find $f(-0.5)$ and $f(2)$	7	L3	CO1
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x	-2	0	1	3	4
f(x)	-23	1	4	82	193

	c.	Find the Polynomial $f(x)$ using Lagrange's Interpolation formula given	7	L3	CO1
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x	0	1	2	5
f(x)	2	3	12	147

Module - 4

Q.7	a.	Solve $\cos x (e^y + 1)dx + \sin x e^y dy = 0$	6	L3	CO2
	b.	Solve $\frac{d^2y}{dx^2} - 4y = \cosh(2x - 1) + 3x$	7	L3	CO2
	c.	Solve $(D^2 + 3D + 2)y = 1 + 3x + x^2$	7	L3	CO2

OR

Q.8	a.	Solve $\frac{dy}{dx} + \frac{y}{x} = x^2 y^6$	6	L3	CO2
	b.	Solve $\left(3x^2 y + \frac{y}{x}\right)dx + (x^3 + \log_e x)dy = 0$	7	L3	CO2
	c.	Solve $(D^2 - 4)y = e^x + \cos 2x$	7	L3	CO2

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Module - 5

Q.9	a.	Using Simpson's one third rule to find $\int_0^{0.6} e^{-x^2} dx$ by taking seven ordinates.	6	L3	CO2
	b.	Evaluate $y(1.4)$ by Milne's method, for the differential equation, $\frac{dy}{dx} = x^2(1+y)$ given that $y(1) = 1$, $y(1.1) = 1.233$, $y(1.2) = 1.548$ and $y(1.3) = 1.979$.	7	L3	CO2
	c.	Find the solution of $\frac{dy}{dx} = x^2y - 1$, by Taylor's series method at $x = 0.2$ considering up to 4 th degree terms, given that $y(0) = 1$.	7	L3	CO2
OR					
Q.10	a.	Using Runge Kutta method of 4 th order solve $\frac{dy}{dx} = \frac{(y^2 - x^2)}{(y^2 + x^2)}$ with $y(0) = 1$ at $x = 0.2$.	6	L3	CO2
	b.	Applying Modified Euler's method, find $y(0.2)$ by solving the equation $\frac{dy}{dx} = x - y^2$, with $y(0) = 1$ taking $h = 0.1$	7	L3	CO2
	c.	Evaluate $\int_0^1 \frac{x}{1+x^2} dx$ using Weddle's rule ($n = 6$)	7	L3	CO2



Second Semester B.E./B.Tech. Degree Examination, June/July 2026
Numerical Methods : CSE Stream

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. M : Marks, L: Bloom's level, C: Course outcomes.
 3. VTU Formula Hand Book is permitted.*

Module – 1			M	L	C
Q.1	a.	Round off the number 865250 to four significant figures and compute absolute error, relative error and percentage error.	6	L2	CO1
	b.	Find a real root of the equation $xe^x - 2 = 0$ correct to three decimal places using Newton-Raphson method. (carryout 3 iterations)	7	L3	CO1
	c.	Find a root of $x^3 - 4x + 1 = 0$ using Bisection method. (carryout 5 iterations).	7	L3	CO1
OR					
Q.2	a.	Find the absolute error if the number $X = 0.00545828$ is i) Truncated to 3 decimal digits ii) Rounded off to 3 decimal digits	6	L2	CO1
	b.	Find a real root of the equation $\cos x - 3x + 1 = 0$ by using Regula-Falsi method.	7	L3	CO1
	c.	Using Secant method, find a real root of the equation $x^3 - 2x + 5 = 0$, correct to three decimal places. (carryout 4 iterations).	7	L3	CO1
Module – 2					
Q.3	a.	If $V = (0, -1, 1, 2)$ and $W = (1, 5, 1, 8)$ compute L_1, L_2, L_∞ norms of V, W and $2V+W$.	6	L2	CO1
	b.	Find the dominant Eigen value and the corresponding Eigen vector of the matrix $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ by power method taking the initial Eigen vector as $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.	7	L3	CO1
	c.	Using Jacobi's Method to find the Eigen values and the corresponding Eigen vectors of the matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$	7	L3	CO1
1 of 3					

OR

Q.4	a.	Show that the system of equations ill conditioned $x + 3y = 4$, $2x + 5y + 7$.	6	L2	CO1
	b.	Apply Gauss-Seidel iteration method upto 3 iteration to solve $10x + y + z = 12$, $x + 10y + z = 12$, $x + y + 10z = 12$	7	L3	CO1
	c.	Use LU de composition method to solve the following system of equations $2x + y + 4z = 12$, $4x + 11y - z = 33$, $8x - 3y + 2z = 20$.	7	L3	CO1

Module - 3

Q.5	a.	The area of a circle (A) corresponding to diameter (D) is given below <table><tr><td>D</td><td>80</td><td>85</td><td>90</td><td>95</td><td>100</td></tr><tr><td>A</td><td>5026</td><td>5674</td><td>6362</td><td>7088</td><td>7854</td></tr></table> Find the area corresponding to diameter 105 using an interpolation formula.	D	80	85	90	95	100	A	5026	5674	6362	7088	7854	7	L3	CO1
D	80	85	90	95	100												
A	5026	5674	6362	7088	7854												
	b.	Use Lagrange's interpolation formula to find $f(4)$ given <table><tr><td>x</td><td>0</td><td>2</td><td>3</td><td>6</td></tr><tr><td>f(x)</td><td>-4</td><td>2</td><td>14</td><td>158</td></tr></table>	x	0	2	3	6	f(x)	-4	2	14	158	7	L3	CO1		
x	0	2	3	6													
f(x)	-4	2	14	158													
	c.	Obtain the Piecewise linear interpolating polynomial defined by the data. <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>f(x)</td><td>1</td><td>2</td><td>5</td><td>10</td></tr></table> And hence find the value of y at $x = 0.5$ & 1.5.	x	0	1	2	3	f(x)	1	2	5	10	6	L3	CO1		
x	0	1	2	3													
f(x)	1	2	5	10													

OR

Q.6	a.	From the following table find the number of students who obtained marks between 40 and 45 <table border="1"> <tr> <td>Marks</td><td>30-40</td><td>40-50</td><td>50-60</td><td>60-70</td><td>70-80</td></tr> <tr> <td>No. of students</td><td>31</td><td>42</td><td>51</td><td>35</td><td>31</td></tr> </table>	Marks	30-40	40-50	50-60	60-70	70-80	No. of students	31	42	51	35	31	7	L3	CO1
Marks	30-40	40-50	50-60	60-70	70-80												
No. of students	31	42	51	35	31												
	b.	Given $u_0 = -5$, $u_1 = -14$, $u_4 = -125$, $u_8 = -21$, $u_{10} = 355$ by using Newton's divided difference formula to find u_2 .	7	L3	CO1												
	c.	Obtain Piecewise interpolating polynomial and hence find $f(2.0)$ <table border="1"> <tr> <td>x</td><td>-2</td><td>0</td><td>1</td><td>3</td><td>4</td></tr> <tr> <td>y</td><td>-23</td><td>1</td><td>4</td><td>82</td><td>193</td></tr> </table>	x	-2	0	1	3	4	y	-23	1	4	82	193	6	L3	CO1
x	-2	0	1	3	4												
y	-23	1	4	82	193												

Module - 4

Q.7	a.	Solve $(y^3 - 3x^2y)dx - (x^3 - 3xy^2)dy = 0$	6	L3	CO2
	b.	Solve $(4D^4 - 4D^3 - 23D^2 + 12D + 36)y = 0$	7	L3	CO2
	c.	Solve $(D^3 - D^2 + 4D - 4)y = \sinh(2x + 3)$	7	L3	CO2

OR

Q.8	a.	Solve $\frac{dy}{dx} - \frac{1}{2}\left(1 + \frac{1}{x}\right)y + \frac{3y^3}{x} = 0$	6	L3	CO2
	b.	Solve $(x^2 + y^2 + x)dx + xydy = 0$	7	L3	CO2
	c.	Solve $(D^3 - 1)y = 3\cos 2x$	7	L3	CO2

Module - 5

Q.9	a.	Use Simpson's $\frac{1}{3}$ rd rule with six equal parts to evaluate $\int_0^6 3x^2 dx$.	6	L3	CO1
	b.	Using Runge-Kutta method of 4 th order to find $y(0.2)$ for the equation $\frac{dy}{dx} = \frac{y-x}{y+x}$, $y(0) = 1$ taking $h = 0.2$.	7	L3	CO1
	c.	Apply Milne's method to compute $y(1.4)$ correct to 4 decimal places given $\frac{dy}{dx} = x^2 + \frac{y}{2}$ and the following data $y(1) = 2$, $y(1.1) = 2.2156$, $y(1.2) = 2.4649$, $y(1.3) = 2.7514$.	7	L3	CO1

OR

Q.10	a.	By using Weddle's rule with 6 equal strips, find the value of $\int_0^1 \frac{x}{1+x^2} dx$	6	L3	CO1
	b.	Apply Taylor's series method to find y at $x = 0.1$ given $\frac{dy}{dx} = 2y + 3e^x$ & $y(0) = 0$.	7	L3	CO1
	c.	Using modified Euler's method find $y(20.2)$ & $y(20.4)$ given that $\frac{dy}{dx} = \log_{10}\left(\frac{x}{y}\right)$ with $y(20) = 5$ taking $h = 0.2$.	7	L3	CO1
